

SIMATS ENGINEERING

CO1 - ASSESSMENT TOOL 1

Analytical Problem Solving

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

FACULTY : Dr. D.SUDHAGAR

COURSE OUTCOME COVERED

CO 1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

CO1 Weightage: 15%

Assessment Tool Weightage: 35%

Duration: 30 minutes

Marks: 25

Code of Conduct:

I K Gowthami(192525428) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K Gowthami

QUESTIONS:05

Total Marks:25

1. Learning Algorithms (Optimization Behavior)

Answer

A high initial learning rate allows the model to make large weight updates and quickly move toward the minimum of the loss function. As the learning rate is gradually decreased (learning rate decay), the updates become smaller, allowing the model to converge more accurately.

Effects of High Learning Rate with Decay

- Fast initial convergence.
- Escapes shallow local minima and plateaus.
- Fine-tunes the solution with smaller updates later.

When it Outperforms a Constant Learning Rate

This strategy performs better when:

- The loss function is smooth or convex.
- The initial learning rate is not excessively large.
- Learning rate decay is applied gradually.
- The dataset is large and complex.

Possible Risks

- If the initial learning rate is too high, the model may overshoot the minimum.
- Very rapid decay may stop learning too early.
- Very slow decay may delay convergence.

Conclusion

Learning rate decay combines fast learning in the beginning with stable convergence at the end, often giving better performance than a constant learning rate.

2. Maximum Likelihood Estimation (MLE)

Derivation of MLE for Mean (μ)

For a Gaussian distribution with known variance:

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i$$

Thus, the Maximum Likelihood Estimator of the mean is simply the sample mean.

Effect of Outliers

- Outliers significantly change the sample mean.
- Even one extreme value can shift the estimate.
- Therefore, MLE is sensitive to outliers.

Effect of Increasing Sample Size

- As the number of samples increases, the estimate becomes closer to the true mean.
- Estimation variance decreases.
- The estimator becomes more accurate and stable.

Robustness of MLE

- MLE is not robust to outliers.
- It performs well when data truly follows a Gaussian distribution without extreme observations.

3. Building a Machine Learning Algorithm (Model Selection)

Challenges

- Very high dimensionality (10,000 features).
- Small number of training samples (500).
- High risk of overfitting.
- Increased computational cost.
- Presence of irrelevant and redundant features.

Proposed Algorithm Pipeline

1. Data Preprocessing

- Handle missing values.
- Normalize features.

2. Feature Selection

- Remove irrelevant features using statistical tests or LASSO.

3. Dimensionality Reduction

- Apply PCA to reduce feature dimensions.

4. Regularization

- Use L1 or L2 regularization.

5. Model Training

- Train a classifier such as Logistic Regression or SVM.

6. Cross Validation

- Evaluate the model using k-fold cross-validation.

How Overfitting is Reduced

- Feature selection removes unnecessary variables.
- PCA reduces data complexity.
- Regularization penalizes overly complex models.
- Cross-validation improves generalization.

4. Neural Networks (Multilayer Perceptron)

Why More Hidden Neurons Increase Representational Power

Each hidden neuron learns different features of the input data. Adding more neurons allows the network to learn:

- More complex patterns.
- Non-linear relationships.
- Better feature representations.

Thus, the network can approximate highly complex functions.

Why More Neurons Do Not Always Improve Performance

- Large networks may memorize training data.
- Overfitting increases.
- Generalization on unseen data decreases.
- Training becomes slower.
- More computational resources are required.

Overfitting and Generalization

- Overfitting: Excellent performance on training data but poor performance on testing data.
- Generalization: Ability to perform well on new, unseen data.

A balanced network size provides the best generalization.

5. Backpropagation, SGD & Curse of Dimensionality

Impact on Gradient Updates

- High-dimensional data increases the number of parameters.
- Gradients become noisy and unstable.
- Training becomes more difficult.

Impact on SGD Convergence

- SGD requires more iterations.
- Updates become less efficient.
- Loss fluctuates more frequently.

Impact on Model Generalization

- Increased risk of overfitting.
- Sparse data makes learning difficult.
- Poor prediction on unseen data.

Techniques to Overcome These Issues

1. Dimensionality Reduction (PCA)

- Removes redundant features.
- Reduces computational cost.
- Improves training speed.

2. Regularization (L1/L2, Dropout)

- Prevents overfitting.
- Improves generalization.
- Reduces model complexity.

Additional Techniques

- Batch Normalization.
- Early Stopping.
- Feature Selection.
- Data Augmentation.

Conclusion

The curse of dimensionality negatively affects gradient computation, SGD convergence, and model generalization. Applying dimensionality reduction, regularization, and proper optimization techniques helps improve convergence speed, reduce overfitting, and achieve better performance on unseen data.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand